Document 2: Literature Review & Evaluation

Cosmetics Store Analytics & Business Intelligence System

Document: 02 - Literature Review **Team Name:** Team 5

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1. Introduction

Retail stores operate in highly competitive environments where effective inventory management and sales monitoring directly impact business survival. In retail cosmetics stores, products range from everyday skincare items to premium perfumes and seasonal makeup. Managing hundreds of distinct items, fluctuating supplier costs, customer discounts, and multiple sales channels presents significant operational challenges. Traditionally, small retail businesses relied on manual bookkeeping or simple spreadsheets. However, as transaction volumes grow, manual methods become slow, error-prone, and incapable of providing timely business insights. This project explores how standard data analytics and business intelligence methods can be applied to retail transaction records to improve operational visibility and decision-making.

2. Review of Relevant Concepts

2.1 Relational Database Management Systems (RDBMS) & SQL

A relational database organizes structured data into tables consisting of rows and columns. Relational Database Management Systems (such as Microsoft SQL Server) enforce data integrity through primary keys, foreign keys, unique constraints, and check constraints. Database normalization divides large, repetitive tables into smaller, related tables to eliminate redundant data and prevent update anomalies. Structured Query Language (SQL) allows analysts to query large transaction volumes, join related records, compute aggregates, and maintain reliable audit trails across retail operations.

2.2 Data Cleaning and ETL (Extract, Transform, Load) Processes

In real-world business environments, raw transaction data is rarely clean. It often contains duplicated records, missing customer identifiers, negative quantities, and formatting inconsistencies across dates and numbers. ETL is the foundational data engineering pipeline where data is extracted from source systems, transformed through validation and cleaning rules, and loaded into an analytical database. Using Python libraries such as pandas, analysts can automate data verification, reconcile line items with invoice headers, and filter out invalid transactions before loading the data into SQL Server.

2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the critical stage of investigating datasets to discover patterns, detect anomalies, test hypotheses, and verify assumptions using descriptive statistics and graphical representations. In retail analytics, EDA helps identify top-selling product categories, measure average transaction sizes, observe hourly sales distributions, and evaluate inventory valuation. By examining both Pearson and Spearman correlations, analysts can assess relationships between transaction metrics while understanding the influence of occasional large commercial orders.

2.4 Business Intelligence (BI) & Data Visualization

Business Intelligence refers to strategies and technologies used by enterprises for data analysis and business information reporting. Effective data visualization translates complex numerical tables into intuitive charts such as monthly revenue trends, department breakdowns, and profit distribution histograms. Visual dashboards enable store owners and operational staff to quickly evaluate performance without needing to write technical database queries.

3. Feedback & Evaluation

Formal supervisor review pending submission.

Because this project is an academic coursework assignment, formal feedback and evaluation comments from the course instructor or teaching assistant will be received following the submission of the initial deliverables. The team will document all supervisor feedback and incorporate required recommendations into the final project revision.

4. Suggested Improvements

Based on our evaluation of current retail analytics practices, several realistic improvements could be added to extend the system:

- **Automated Data Ingestion Pipeline:** Set up an automated script that monitors a designated folder for new daily transaction files, runs cleaning checks automatically, and appends valid records directly to the database.
- **Low-Stock & Discrepancy Alerts:** Implement an alert mechanism that flags products with negative inventory quantities or stock falling below minimum safety thresholds to support timely reordering.
- **Customer Loyalty Program Tracking:** Introduce customer loyalty account numbers for walk-in retail shoppers to track repeat purchases and measure customer retention over time.
- **Basic Demand Forecasting:** Use simple moving averages or basic time-series models to predict expected product demand for upcoming retail seasons based on historical sales patterns.

5. Final Grading Criteria

The project will be evaluated according to the standard academic assessment categories specified in the assignment. In accordance with project guidelines, no specific marks or percentage weights are invented:

- **Documentation:** Clarity, completeness, and academic structure of all written deliverables, including project planning, literature review, and requirements specifications.
- **Implementation:** Correctness and quality of the data preprocessing pipeline, relational schema design, SQL scripts, and Python exploratory analysis code.
- **Testing:** Verification of data consistency, mathematical reconciliation between invoices and line items, and enforcement of relational database constraints.
- **Presentation:** Ability of Team 5 members to explain the project methodology, demonstrate system outputs clearly, and answer questions during the final academic discussion.